

NLP

Week#01

Assessment Methods

Assessment		Percentage
1	Quizzes (10-15%)	10%
2	Assignments (5-10%)	10%
3	Mid-Term Exam	20%
4	End-Semester Exam	40%
5	Project	10%
6	Labs	10%
Total		100%

We assume you have learned the following concepts already:

- Language models
- Logistic regression w/ regularization
- Unsupervised vs supervised learning
- Feedforward neural networks, convolutional neural networks
- PyTorch programming • (A little bit of reinforcement learning)

What is NLP?

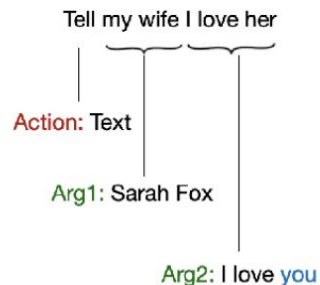
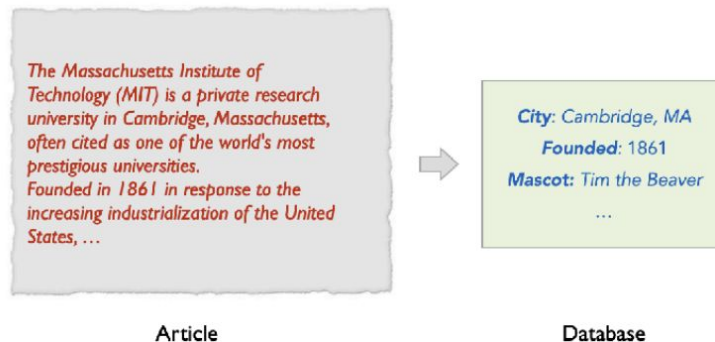
Natural Language Processing

- NLP = building **computer programs** to analyze, understand and generate **human language** - either spoken or written (informal)
- NLP is an interdisciplinary field



NLP subfields

- Machine translation
- Question answering
- Information extraction
- Text summarization
- Dialogue
- Semantic parsing
- Syntactic parsing
- Language grounding
-



Text Mining

Text mining, also known as text data mining, is the process of transforming unstructured text into a structured format to identify meaningful patterns and new insights. You can use text mining to analyze vast collections of textual materials to capture key concepts, trends and hidden relationships.

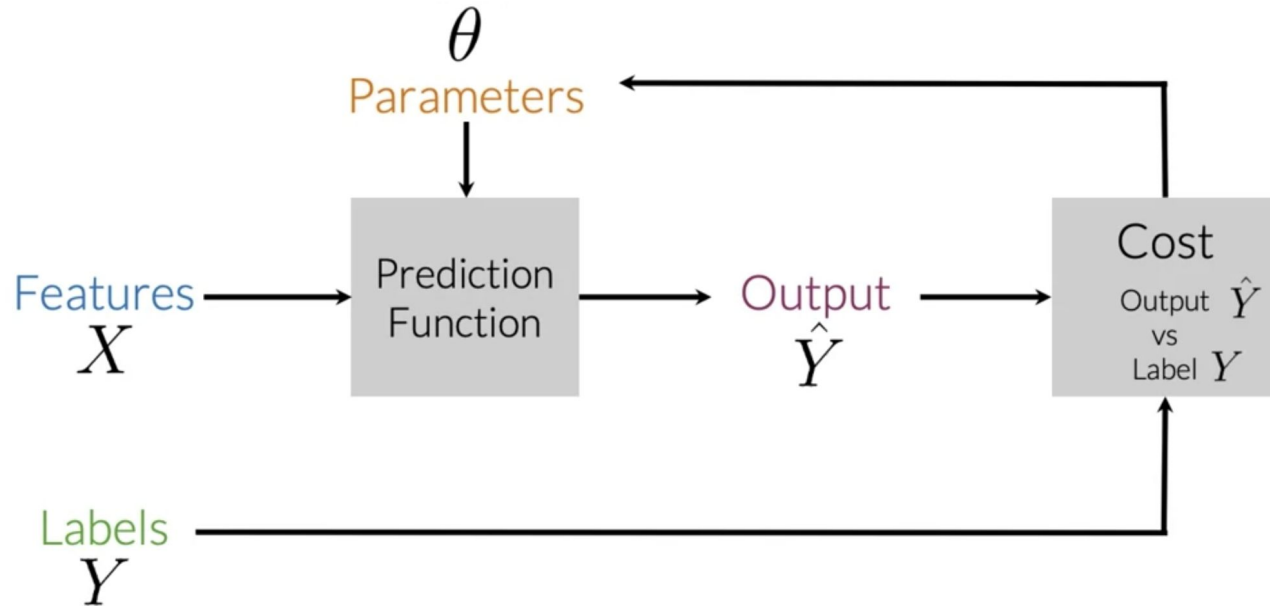
Text mining is a part of Data mining to extract valuable text information from a text database repository. Text mining is a multi-disciplinary field based on data recovery, Data mining, [AI, statistics](#), [Machine learning](#), and computational linguistics.

Text mining helps us turn that raw text into structured, analyzable data.

Supervised ML & Sentiment Analysis

Supervised ML(training)

In supervised machine learning, you usually have an input X , which goes into your prediction function to get your $\hat{Y}$. You can then compare your prediction with the true value Y . This gives you your cost which you use to update the parameters θ . The following image, summarizes the process.



Sentiment Analysis

“I hate being stuck in traffic for hours.” → Negative

“The service at this restaurant was excellent.” → Positive



Tweet: I am happy, because I am learning NLP.

POSITIVE

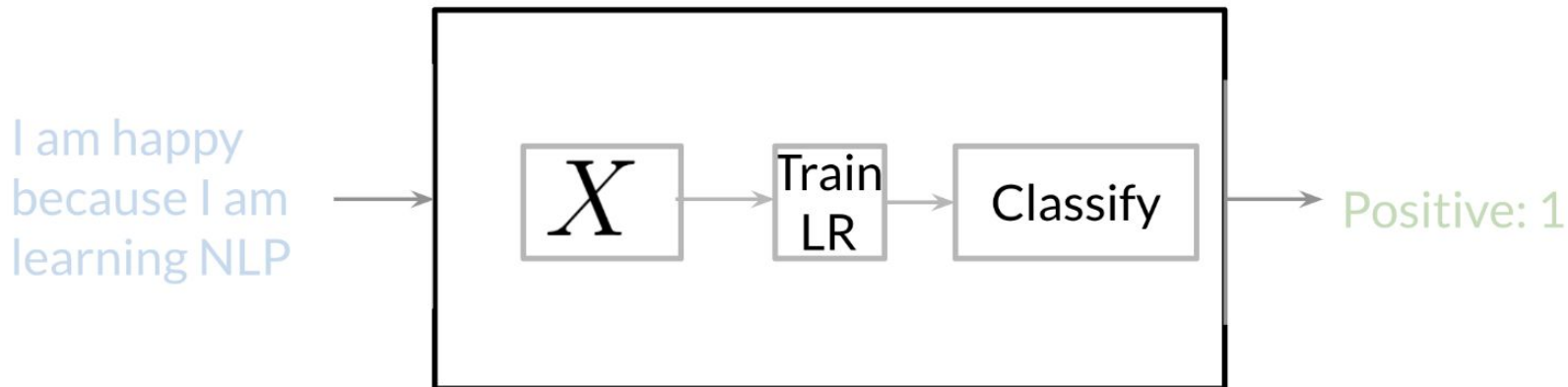


Tweet: I am sad, because I am not learning NLP.

NEGATIVE

Supervised ML - Sentiment Analysis

To perform sentiment analysis on a tweet, you first have to represent the text (i.e. "I am happy because I am learning NLP ") as features, you then train your logistic regression classifier, and then you can use it to classify the text.



Note that in this case, you either classify 1, for a positive sentiment, or 0, for a negative sentiment.

Logistic Regression

Logistic regression is a type of supervised learning classification algorithm that is adept not only in binary classification but also in multinomial classification problems. It can predict whether an instance belongs to one of multiple classes.

How does Logistic Regression work?

In logistic regression, the core operation is transforming input data into a probability value, which is achieved using the sigmoid function.

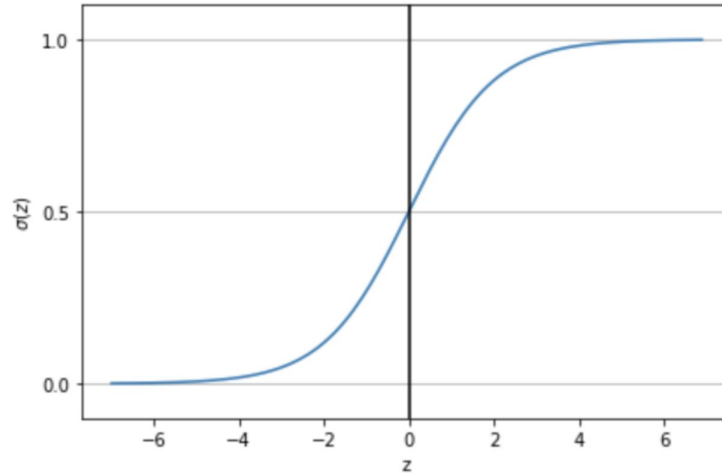
logistic regression is used when the output is categorical. The output in logistic regression needs to represent a probability that a sample belongs to each of the classes.

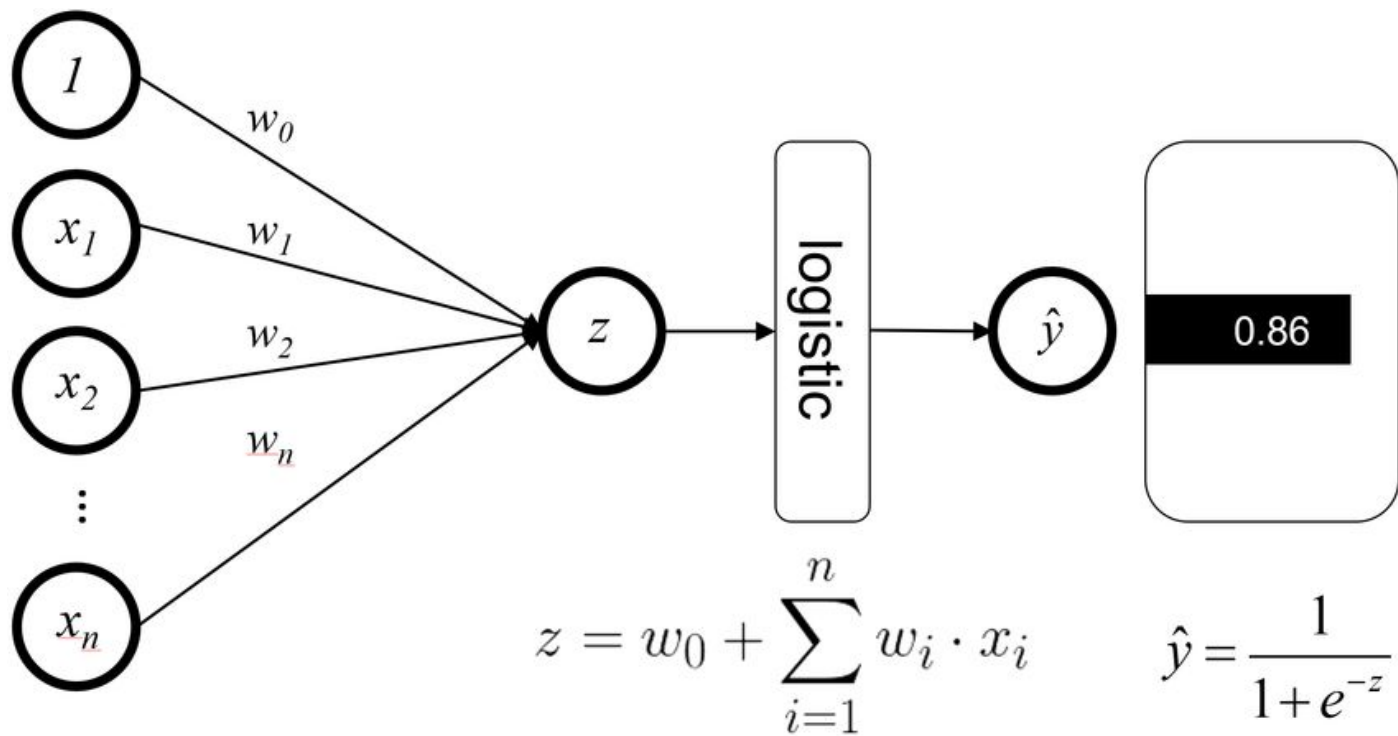
Sigmoid/Logistic Function

The sigmoid function, also known as the logistic function, is a special mathematical equation that transforms any real-valued number into a value between 0 and 1. It's given by:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The sigmoid function, plotted (below) against its input z , illustrates how smoothly transitions from 0 to 1. As z increases, it approaches 1, and as z decreases towards negative infinity, $\sigma(z)$ gets closer to 0. Notably, at $z=0$, $\sigma(z)$ is exactly 0.5.





- **Original cost function (without regularization):**

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log(h_{\theta}(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)})) \right]$$

- **With regularization (penalty term added):**

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log(h_{\theta}(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)})) \right] + \lambda \cdot \text{Penalty}$$

Here:

- λ = regularization strength (higher $\rightarrow$ stronger penalty).

Example

Let's have an example:

We have set of tweets

Let's create vocabulary first

Vocabulary

Tweets:

[tweet_1, tweet_2, ..., tweet_m]



I am happy because I am learning NLP

...

...

...

I hated the movie

$V =$

[I, am, happy, because, learning, NLP, ... hated, the, movie]

Vocabulary & Feature Extraction

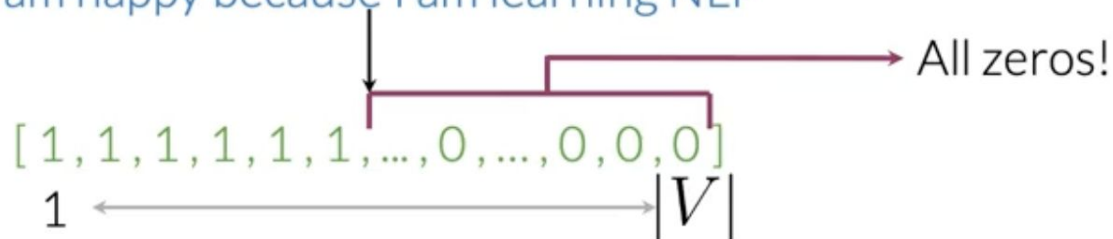
I am happy because I am learning NLP

[I, am, happy, because, learning, NLP, ... hated, the, movie]

↓	↓	↓	↓	↓	↓	↓	↓	↓	↓
[1, 1,	1,	1,	1,	1,	1, ...	0,	0,	0]	

Problems with sparse representations

I am happy because I am learning NLP



Coding Example - Product Reviews

<https://colab.research.google.com/drive/17M3Rj-kQtGr1Ymht-ARIL6UPmj17mqST?usp=sharing>

- **Precision:** Measures the accuracy of positive predictions. It answers the question, “Of all the items the model labeled as positive, how many were actually positive?”
- **Recall (Sensitivity):** Measures the model’s ability to find all the positive instances. It answers the question, “Of all the actual positives, how many did the model correctly identify?”
- **F1 Score:** The harmonic mean of precision and recall. It balances the two metrics into a single number, making it especially useful when precision and recall are in trade-off.

Precision, Recall, and F1 Score Formulas

1. Precision:

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}$$

- Focuses on the correctness of positive predictions.

2. Recall:

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

- Measures how well the model captures all positive instances.

3. F1 Score:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- A single metric that balances precision and recall.

Confusion Matrix

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	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

Real-Life Example: Precision and Recall Trade-offs

- **Scenario 1: Email Spam Classifier**
 - Precision matters more: Minimizing false positives is critical to avoid classifying important emails as spam.
- **Scenario 2: Cancer Detection**
 - Recall matters more: Minimizing false negatives ensures that no cases of cancer go undetected.
- **Trade-Offs**
 - Increasing precision may decrease recall and vice versa. The F1 score balances this trade-off.

- If the filter only marks obvious spam → High precision, low recall.
 - If the filter marks everything suspicious → High recall, low precision.
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- A strict test (only positive if virus level is very high) → High precision, low recall.
 - A sensitive test (flags even small traces) → High recall, low precision.
-
- Quick Analogy: Fishing 🎣
 - Precision: Of the fish you caught, how many are really tuna (the fish you wanted)?
 - Recall: Out of all tuna in the sea, how many did you manage to catch?

Key Takeaways

Choose the Right Metric:

- Use precision for problems where false positives are critical.
- Use recall for problems where false negatives are critical.
- Use F1 score when precision and recall are equally important.

Task

<https://www.kaggle.com/code/abdmental01/text-preprocessing-nlp-steps-to-process-text>

Task

```
corpus = [  
    'This is the first document',  
    'This document is the second document',  
    'and this is the third one',  
    'is this the first document'  
]
```

The goal of this task is to build and evaluate a machine learning model that can automatically predict the **sentiment (positive or negative)** of Amazon product reviews.